Sumanth Rao

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Ph.D. Candidate, Computer Science • University of California, San Diego

EDUCATION

University of California, San Diego 2021 – 2027 (expected)
Ph.D. in Computer Science • Advisors: Stefan Savage and Geoffrey M. Voelker • Sysnet & CryptoSec groups

PES University, Bengaluru, India 2016 – 2020
B.Tech. in Computer Science • First Class with Honors • GPA 9.6/10

EXPERIENCE

Google — Gmail Delivery Security | *Student Researcher* Jun 2026 – Sep 2026
Mentor: Weihaw Chaung

- Built an observatory that detects DNS-based domain hijacks attackers use to spoof SPF authentication.
- Deployed to Google prod (joining WHOIS, passive DNS), where it scans 200M Gmail domains every day.

Corelight Labs | *Research Intern* Jun 2025 – Sep 2025
Mentor: Yacin Nadji

- Built Zeek detection for DCSync/DCShadow-style Active Directory attacks in east/west customer traffic.
- Fingerprinted DRSUAPI replication over DCE-RPC to separate legitimate DCs from rogue replicators.

Google — Virtual Cluster | *Software Engineering Intern* Jun 2022 – Sep 2022

- Reduced web-proxying overhead by 7× inside Virtual Cluster, Google's sandboxed mock prod environment.

Citrix — Licensing | *Software Engineer I* Aug 2020 – Aug 2021

- Reduced License Server test resource use by up to 50% with Docker microservices on Citrix Xen.
- Earned the highest level of Citrix's internal software security certification.

Zulip Open Source Project | *Volunteer Contributor* 2019 – 2026
Mentor: Tim Abbott • Google Summer of Code (GSoC)

- Helped develop the terminal client (GSoC 2021) and later mentored five GSoC students (2022–2026).

SELECT PUBLICATIONS

Lost in Translation: Text Message Spoofing via Email — Distinguished Paper Award
IEEE S&P (Oakland) 2026 • **First author** • CSAW Applied Research Finalist 2026 • CVE-2025-46311 via fuzzing

Patient Care Technology Disruptions Associated With the CrowdStrike Outage
JAMA Network Open 2025 • **First author** • *Wired*, *Forbes*

Unfiltered: Measuring Cloud-based Email Filtering Bypasses
ACM Web Conference (WWW) 2024 • **First author** • *DarkReading*, ACM TechNews

BLE Theft Auto: Evaluating the Security of Aftermarket BLE-based Automotive Remote Control Systems
USENIX Security 2026 • Co-author • CVE-2026-18411, CISA ICSA-26-216-01 • *Wired* • **Presented at DEF CON 34**

No Privacy Among Spies: Assessing the Functionality and Insecurity of Consumer Android Spyware Apps
Privacy Enhancing Technologies Symposium (PETS) 2023 • Co-author • **Presented at FTC PrivacyCon 2024**

IN SUBMISSION & PREPARATION

Taking the Pulse: Internet Monitoring of Hospital Networks • *In Submission*
Infers hospital outages (ransomware, power loss) from externally visible host and clinical-service availability

Victim as a Service: LLM Honeypot for Engaging with Interactive Scammers • *In Submission*
An interactive honeypot on Bluesky to study what draws scammers to targets and how conversational scams unfold.

TECHNICAL SKILLS

Languages Python, C++, Rust, Go, Java, JavaScript, Shell, SQL.
Internet Measurement Censys, PassiveDNS, ZMap, ZDNS, RIPE Atlas.
Network Analysis Zeek, Wireshark, Mitmproxy, Traffic fingerprinting.
Reverse Engineering Ghidra, JADX, Android app and firmware analysis, BLE protocol analysis.
AI Security LLM jailbreaks, prompt-injection attacks & defenses, Cyber ranges.
Systems & Infrastructure Linux, Docker, Git, Django.

DISCLOSURES, PATENTS & MEDIA COVERAGE

- **CVE-2026-18411** — KARR Security System auth bypass; disclosed via CISA (ICSA-26-216-01).
- **CVE-2025-46311** — iOS message sender-parsing flaw enabling spoofing of arbitrary phone numbers.
- **Provisional U.S. Patent 63/674,158**, "Digital Availability Surveillance Method", assigned to UCSD (2024).
- Research covered by Wired, Forbes, DarkReading, ACM TechNews, CACM News, KPBS, CTV News.

AWARDS, HONORS & SERVICE

- **2026** Distinguished Paper Award, IEEE Symposium on Security and Privacy (Oakland).
- **2026** CSE Doctoral Award for Excellence in Service and Leadership, UC San Diego.
- **2026** Finalist, CSAW Applied Research Competition.
- **2025** Top 5, BSides San Diego CTF.
- **2024** Finalist, NatSec Hackathon (Stanford DEFCON & Shield Capital).
- **2021–2022** JN Tata Endowment Scholar (national scholarship for graduate study abroad).
- **Program Committee**: SIGMETRICS 2027, IEEE EuroS&P 2027, RAID 2026.